

ON DIRICHLET SERIES FOR SUMS OF SQUARES

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ABSTRACT. In [12], Hardy and Wright recorded elegant closed forms for the generating functions of the divisor functions $\sigma_k(n)$ and $\sigma_k^2(n)$:

$$\sum_{n=1}^{\infty} \frac{\sigma_k(n)}{n^s} = \zeta(s)\zeta(s-k)$$

and

$$\sum_{n=1}^{\infty} \frac{\sigma_k^2(n)}{n^s} = \frac{\zeta(s)\zeta(s-k)^2\zeta(s-2k)}{\zeta(2s-2k)}.$$

In this paper, we explore other arithmetical functions enjoying this remarkable property. In Theorem 2.1 below, we are able to generalize the above result and prove that if f_i and g_i are completely multiplicative, then we have

$$\sum_{n=1}^{\infty} \frac{(f_1 * g_1)(n) \cdot (f_2 * g_2)(n)}{n^s} = \frac{L_{f_1 f_2}(s) L_{g_1 g_2}(s) L_{f_1 g_2}(s) L_{g_1 f_2}(s)}{L_{f_1 f_2 g_1 g_2}(2s)}$$

where $L_f(s) := \sum_{n=1}^{\infty} f(n)n^{-s}$ is the Dirichlet series corresponding to f . Let $r_N(n)$ be the number of solutions of $x_1^2 + \cdots + x_N^2 = n$ and $r_{2,P}(n)$ be the number of solutions of $x^2 + Py^2 = n$. One of the applications of Theorem 2.1 is to obtain closed forms, in terms of $\zeta(s)$ and Dirichlet L -functions, for the generating functions of $r_N(n)$, $r_N^2(n)$, $r_{2,P}(n)$ and $r_{2,P}(n)^2$ for certain N and P . We also use these generating functions to obtain asymptotic estimates of the average values for each function for which we obtain a Dirichlet series.

1. INTRODUCTION

Let σ_k denote the sum of k th powers of the divisors of n . It is also quite usual to write d for σ_0 and τ for σ_1 . There is a beautiful formula for the generating functions of $\sigma_k(n)$ (see Theorem 291 in Chapter XVII of [12])

$$(1.1) \quad \sum_{n=1}^{\infty} \frac{\sigma_k(n)}{n^s} = \zeta(s)\zeta(s-k), \quad \Re(s) > \max\{1, k+1\}$$

which is in terms of only the Riemann Zeta function $\zeta(s)$. Following Hardy and Wright, by standard techniques, one can prove the following remarkable identity (see Theorem 305 in Chapter XVII of [12])

$$(1.2) \quad \sum_{n=1}^{\infty} \frac{\sigma_a(n)\sigma_b(n)}{n^s} = \frac{\zeta(s)\zeta(s-a)\zeta(s-b)\zeta(s-a-b)}{\zeta(2s-a-b)}$$

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for $\Re(s) > \max\{1, a+1, b+1, a+b+1\}$. In this paper, we identify other arithmetical functions enjoying similarly explicit representations. In Theorem 2.1 of §2 below, we are able to generalize the above result and prove that if f_i and g_i are completely multiplicative, then we have

$$\sum_{n=1}^{\infty} \frac{(f_1 * g_1)(n) \cdot (f_2 * g_2)(n)}{n^s} = \frac{L_{f_1 f_2}(s) L_{g_1 g_2}(s) L_{f_1 g_2}(s) L_{g_1 f_2}(s)}{L_{f_1 f_2 g_1 g_2}(2s)}$$

where $L_f(s) := \sum_{n=1}^{\infty} f(n)n^{-s}$ is the Dirichlet series corresponding to f . As we shall see, this result recovers Hardy and Wright's formulae (1.1) and (1.2) immediately.

More generally, for certain classes of Dirichlet series, $\sum_{n=1}^{\infty} A(n)n^{-s}$, our Theorem 2.1 can be applied to obtain closed forms for the series $\sum_{n=1}^{\infty} A^2(n)n^{-s}$. In particular, if the generating function $L_f(s)$ of an arithmetic function f is expressible as a sum of products of two L -functions:

$$L_f(s) = \sum_{\chi_1, \chi_2} a(\chi_1, \chi_2) L_{\chi_1}(s) L_{\chi_2}(s)$$

for certain coefficients $a(\chi_1, \chi_2)$ and Dirichlet characters χ_i , then we are able to find a simple closed form (in term of L -functions) for the generating function $L_f^2(s) := \sum_{n=1}^{\infty} f^2(n)n^{-s}$.

One of our central applications is to the study of the number of representations as a sum of squares. Let $r_N(n)$ be the number of representations of n as a N squares. Hardy and Wright record a classical closed form, due to Lorenz, of the generating function for $r_2(n)$ in the terms of $\zeta(s)$ and a Dirichlet L -function, namely,

$$\sum_{n=1}^{\infty} \frac{r_2(n)}{n^s} = 4\zeta(s) L_{-4}(s)$$

where $L_{\mu}(s) = \sum_{n=1}^{\infty} \left(\frac{\mu}{n}\right) n^{-s}$ is the *primitive L -function* corresponding to the *Kronecker symbol* $\left(\frac{\mu}{n}\right)$. Define

$$L_N(s) := \sum_{n=1}^{\infty} \frac{r_N(n)}{n^s} \quad \text{and} \quad R_N(s) := \sum_{n=1}^{\infty} \frac{r_N^2(n)}{n^s}.$$

Simple closed forms for $L_N(s)$ are known for $N = 2, 4, 6$ and 8 ; indeed the corresponding q -series were known to Jacobi. The entity $L_3(s)$ in particular is still shrouded in mystery, as a series relevant to the study of lattice sums in the physical sciences. Lately there has appeared a connection between L_3 and a modern theta-cubed identity of G. Andrews [1] which we list in (6.7), R. Crandall [4] and p. 301 of [2]. In §3, we shall obtain simple closed forms for $R_N(s)$ for these N from the corresponding $L_N(s)$, via Theorem 2.1. Since the generating functions are accessible, by evaluating the residues of their largest real poles and Cauchy's integral theorem, we are also able to deduce

$$\sum_{n \leq x} r_N^2(n) = W_N x^{N-1} + O(x^{N-5/3+\epsilon})$$

for $N = 4, 6, 8$ where

$$(1.3) \quad W_N := \frac{1}{(N-1)(1-2^{-N})} \frac{\pi^N}{\Gamma^2(\frac{1}{2}N)} \frac{\zeta(N-1)}{\zeta(N)}, \quad (N \geq 3).$$

These results also serve to independently establish some initial cases of a general conjecture due to Wagon:

Wagon's Conjecture. *For $N \geq 3$, $r_N^2(n)$ is (in sum) asymptotic to $W_N x^{N-1}$.*

This conjecture motivated our interest in such series representations. Recently, the conjecture has been proved by Crandall and Wagon in [7]. In fact, it is proved that

$$\lim_{x \rightarrow \infty} x^{1-N} \sum_{n < x} r_N^2(n) = W_N,$$

with various rates of convergence (those authors found the $N = 3$ case especially difficult, with relevant computations revealing very slow convergence to the above limit). In their treatment of the Wagon conjecture and related matters, they needed to evaluate the following Dirichlet series

$$\sum_{n=1}^{\infty} \frac{\phi(n)\sigma_0(n^2)}{n^s}$$

and we have established, by an easier version of what follows, that it is

$$\sum_{n=1}^{\infty} \frac{\phi(n)\sigma_0(n^2)}{n^s} = \zeta^3(s-1) \prod_p \left(1 - \frac{3}{p^s} - \frac{1}{p^{2s-2}} + \frac{4}{p^{2s-1}} - \frac{1}{p^{3s-2}} \right)$$

where the product is over all primes. A word is in order concerning the importance of first- and second-order summatories. In a theoretical work [5] and a computational one [7] it is explained that the Wagon conjecture implies that *sums of three squares have positive density*. This interesting research connection is what inspired Wagon to posit his computationally motivated conjecture. Though it is known that the density of the set $S = \{x^2 + y^2 + z^2\}$ is exactly $5/6$ (e.g [10]), there are intriguing signal-processing and analytic notions that lead more easily at least to positivity of said density. Briefly, the summatory connection runs as follows. From the Cauchy-Schwarz inequality we know

$$\#\{n < x; n \in S\} > \frac{(\sum_{n < x} r_3(n))^2}{\sum_{n < x} r_3^2(n)},$$

so the Wagon conjecture even gives an explicit numerical lower bound on the density of S . Of course, the density for sums of more than 3 squares is likewise positive, and boundable, yet the Lagrange theorem that sums of four squares comprise *all* nonnegative integers dominates in the last analysis. Still, the signal-processing and computational notions of Crandall and Wagon forge an attractive link between these L -series of our current interest and additive number theory.

In §4 and §5, we similarly study the number of representations by a binary quadratic forms. Let $r_{2,P}(n)$ be the number of solutions of the binary quadratic form $x^2 + Py^2 = n$. Define

$$L_{2,P}(s) := \sum_{n=1}^{\infty} \frac{r_{2,P}(n)}{n^s} \quad \text{and} \quad R_{2,P}(s) := \sum_{n=1}^{\infty} \frac{r_{2,P}(n)^2}{n^s}.$$

The closed forms of $L_{2,P}(s)$ has been studied by a number of people, particular by Glasser, Zucker and Robertson (see [9] and [16]). In finding the exact evaluation of lattice sums, they are interested in expressing a multiple sum, such as the generating functions of $r_{2,P}(n)$, as a product of simple sums. As a result, plenty of closed forms

of Dirichlet series $\sum_{(n,m) \neq (0,0)} (am^2 + bmn + cn^2)^{-s}$ in terms of L -functions have been found. One of the most interesting cases is when the binary quadratic forms have *disjoint discriminants*, i.e, have only one form per genus. Then there are simple closed forms for the corresponding $L_{2,P}(s)$ (see (4.1) below). By applying Theorem 2.1, we obtain closed forms for $R_{2,P}(s)$ and from this we also deduce asymptotic estimates for $r_{2,P}(n)$ and $r_{2,P}(n)^2$.

In the last section, we shall discuss $L_N(s)$ for some other less tractable cases. In particular, we collect some representations of the generating function for $r_3(n)$, $r_N(n)$, and discuss $r_{12}(n)$ and $r_{24}(n)$.

Throughout, our notation is consistent with that in [12, 13] and [14]. We should also remark that we were lead to the structures exhibited herein by a significant amount of numeric and symbolic computation: leading to knowledge of the formulae for $R_2, R_4, R_8, R_{2,2}$ and $R_{2,3}$ before finding our general results. And indeed R. Crandall triggered our interest by transmitting his formula for R_4 .

2. BASIC RESULTS

Let $\sigma(f)$ be the *abscissa* of absolute convergence of the Dirichlet series

$$L_f(s) := \sum_{n=1}^{\infty} f(n)n^{-s}.$$

For any two arithmetic functions f and g , define

$$f * g(n) := \sum_{d|n} f(d)g(n/d)$$

to be the *convolution* of f and g .

Theorem 2.1. *Suppose f_1, f_2 and g_1, g_2 are completely multiplicative arithmetic functions. Then for $\Re(s) \geq \max\{\sigma(f_i), \sigma(g_i)\}$, we have*

$$(2.1) \quad \sum_{n=1}^{\infty} \frac{(f_1 * g_1)(n) \cdot (f_2 * g_2)(n)}{n^s} = \frac{L_{f_1 f_2}(s) L_{g_1 g_2}(s) L_{f_1 g_2}(s) L_{g_1 f_2}(s)}{L_{f_1 f_2 g_1 g_2}(2s)}.$$

Proof. Since $(f_1 * g_1)(n) \cdot (f_2 * g_2)(n)$ is multiplicative, we only need to consider its values at the prime powers. For any prime p and any $l \geq 0$,

$$(f_i * g_i)(p^l) = \sum_{d|p^l} f_i(d)g_i(p^l/d) = \frac{f_i(p)^{l+1} - g_i(p)^{l+1}}{f_i(p) - g_i(p)},$$

as each of f_1, f_2, g_1, g_2 is completely multiplicative. We intend above that if both $f_i(p)$ and $g_i(p)$ are zero, then

$$(f_i * g_i)(p^l) = \begin{cases} 1 & \text{if } l = 0; \\ 0 & \text{if } l \geq 1. \end{cases}$$

Thus, we have

$$\begin{aligned}
\Sigma_p &:= \sum_{l=0}^{\infty} (f_1 * g_1)(p^l) (f_2 * g_2)(p^l) p^{-ls} \\
&= \sum_{l=0}^{\infty} \frac{(f_1(p)^{l+1} - g_1(p)^{l+1})(f_2(p)^{l+1} - g_2(p)^{l+1})}{(f_1(p) - g_1(p))(f_2(p) - g_2(p))} p^{-ls} \\
&= \frac{\sum_{l=0}^{\infty} \{ (f_1 f_2)(p)^{l+1} p^{-ls} + (g_1 g_2)(p)^{l+1} p^{-ls} - (f_1 g_2)(p)^{l+1} p^{-ls} - (g_1 f_2)(p)^{l+1} p^{-ls} \}}{(f_1(p) - g_1(p))(f_2(p) - g_2(p))}.
\end{aligned}$$

On summing up all the geometric series, we arrive at

$$\begin{aligned}
\Sigma_p &:= \frac{\frac{(f_1 f_2)(p)}{1 - (f_1 f_2)(p)p^{-s}} + \frac{(g_1 g_2)(p)}{1 - (g_1 g_2)(p)p^{-s}} - \frac{(f_1 g_2)(p)}{1 - (f_1 g_2)(p)p^{-s}} - \frac{(g_1 f_2)(p)}{1 - (g_1 f_2)(p)p^{-s}}}{(f_1(p) - g_1(p))(f_2(p) - g_2(p))} \\
&= \frac{1 - (f_1 f_2 g_1 g_2)(p)p^{-2s}}{(1 - (f_1 f_2)(p)p^{-s})(1 - (g_1 g_2)(p)p^{-s})(1 - (f_1 g_2)(p)p^{-s})(1 - (g_1 f_2)(p)p^{-s})}.
\end{aligned}$$

In view of the Euler product form for a Dirichlet series, we have

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{(f_1 * g_1)(n) \cdot (f_2 * g_2)(n)}{n^s} &= \prod_p \left\{ \sum_{l=0}^{\infty} \frac{(f_1 * g_1)(p^l) (f_2 * g_2)(p^l)}{p^{ls}} \right\} \\
&= \frac{L_{f_1 f_2}(s) L_{g_1 g_2}(s) L_{f_1 g_2}(s) L_{g_1 f_2}(s)}{L_{f_1 f_2 g_1 g_2}(2s)}.
\end{aligned}$$

This proves our theorem. $\square$

A first easy application of Theorem 2.1 is to evaluate the Dirichlet series $\sum_{n=1}^{\infty} \sigma_k(n) n^{-s}$ and $\sum_{n=1}^{\infty} \sigma_a(n) \sigma_b(n) n^{-s}$. If we let $f_1(n) := n^k$, $f_2(n) := \delta(n)$ and $g_1(n) = g_2(n) := 1$ where $\delta(n)$ is 1 if $n = 1$ and 0 otherwise, then

$$\begin{aligned}
L_{f_1 f_2}(s) &= L_{g_1 f_2}(s) = L_{f_1 f_2 g_1 g_2}(s) = 1, \\
L_{f_1 g_2}(s) &= \zeta(s - k), \quad L_{g_1 g_2}(s) = \zeta(s).
\end{aligned}$$

Thus Theorem 2.1 gives the following identity

$$\sum_{n=1}^{\infty} \frac{\sigma_k(n)}{n^s} = \zeta(s) \zeta(s - k)$$

which recovers (1.1). Similarly, if we let $f_1(n) := n^a$, $f_2(n) := n^b$ and $g_1(n) = g_2(n) := 1$, then

$$\begin{aligned}
L_{f_1 f_2}(s) &= L_{f_1 f_2 g_1 g_2}(s) = \zeta(s - (a + b)), \quad L_{g_1 g_2}(s) = \zeta(s), \\
L_{f_1 g_2}(s) &= \zeta(s - a), \quad L_{f_2 g_1}(s) = \zeta(s - b).
\end{aligned}$$

and Theorem 2.1 gives

$$\sum_{n=1}^{\infty} \frac{\sigma_a(n) \sigma_b(n)}{n^s} = \frac{\zeta(s - (a + b)) \zeta(s) \zeta(s - a) \zeta(s - b)}{\zeta(2s - (a + b))}.$$

In particular, for any real λ ,

$$(2.2) \quad \sum_{n=1}^{\infty} \sigma_{\lambda}^2(n) n^{-s} = \frac{\zeta(s - 2\lambda) \zeta(s - \lambda)^2 \zeta(s)}{\zeta(2(s - \lambda))}.$$

We shall discuss more elaborate applications of Theorem 2.1 in the latter sections. Before doing this, we give the following example here to explain why Theorem 2.1 cannot in general be extended nicely to higher order.

We are interested in obtaining the generating functions for the k th moment of $r_2(n)$. For any $n, k \geq 0$, let

$$A(n, k) := \sum_{j=0}^k (-1)^j (k-j)^n \binom{n+1}{j}$$

and

$$E_n(x) := \sum_{k=1}^n A(n, k) x^{k-1}.$$

Then, as is well known, $E_n(x)$ is the n th *Euler polynomial* [3]. Explicitly, the first few Euler polynomials are $E_1(x) = 1$, $E_2(x) = 1 + x$, $E_3(x) = 1 + 4x + x^2$ and $E_4(x) = 1 + 11x + 11x^2 + x^3$. It is also known that $A(n, k)$ satisfies the following recursion: $A(1, 1) = 1$ and

$$A(n, k) = k A(n-1, k) + (n-k+1) A(n-1, k-1).$$

Using this relation, one easily shows that

$$(2.3) \quad \sum_{l=0}^{\infty} l^n x^l = \frac{x E_n(x)}{(1-x)^{n+1}}, \quad n = 1, 2, \dots$$

Equation (2.3) enables us to obtain the generating functions for the higher moments of $r_2(n)$ as follows: for $\mu \equiv 0$ or $1 \pmod{4}$, we let $\left(\frac{\mu}{n}\right)$ be the *Jacobi-Legendre-Kronecker symbol* and again consider

$$L_{\mu}(s) := \sum_{n=1}^{\infty} \left(\frac{\mu}{n}\right) n^{-s}$$

the L -function corresponding to $\left(\frac{\mu}{n}\right)$. It is known (e.g. p. 291 in [2]) that

$$\sum_{n=1}^{\infty} \frac{r_2(n)}{n^s} = 4\zeta(s) L_{-4}(s) = \sum_{n=1}^{\infty} \frac{4(1 * \left(\frac{-4}{n}\right))(n)}{n^s}$$

and $r_2(n) = 4(1 * \left(\frac{-4}{n}\right))(n)$ for any $n \geq 1$. A simple calculation shows that for any $l \geq 0$,

$$\left(1 * \left(\frac{-4}{n}\right)\right)(p^l) = \begin{cases} 1 & \text{if } p = 2; \\ l+1 & \text{if } p \geq 3 \text{ and } \left(\frac{-1}{p}\right) = 1; \\ \frac{(-1)^l + 1}{2} & \text{if } p \geq 3 \text{ and } \left(\frac{-1}{p}\right) = -1. \end{cases}$$

We now have

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{r_2^N(n)}{n^s} &= 4^N \sum_{n=1}^{\infty} \frac{\{(1 * (\frac{-4}{n}))(n)\}^N}{n^s} \\
&= 4^N \prod_p \sum_{l=0}^{\infty} \frac{\{(1 * (\frac{-4}{p^l}))(p^l)\}^N}{p^{ls}} \\
&= \frac{4^N}{1-2^{-s}} \left\{ \prod_{(\frac{-1}{p})=-1} \sum_{l=0}^{\infty} \left(\frac{(-1)^l + 1}{2} \right)^N p^{-ls} \right\} \left\{ \prod_{(\frac{-1}{p})=1} \sum_{l=0}^{\infty} (l+1)^N p^{-ls} \right\} \\
&= \frac{4^N}{1-2^{-s}} \prod_{(\frac{-1}{p})=-1} \frac{1}{1-p^{-2s}} \prod_{(\frac{-1}{p})=1} \frac{E_N(p^{-s})}{(1-p^{-s})^{N+1}}
\end{aligned}$$

on using (2.3). [Here $\prod_p$ denotes the infinite product over all primes.] Firstly, when $N = 2$, we have most pleasingly,

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{r_2^2(n)}{n^s} &= \frac{16}{1-2^{-s}} \prod_{(\frac{-1}{p})=-1} \frac{1}{1-p^{-2s}} \prod_{(\frac{-1}{p})=1} \frac{1+p^{-s}}{(1-p^{-s})^3} \\
&= \frac{16}{1+2^{-s}} \left\{ \frac{1}{1-2^{-s}} \prod_{(\frac{-1}{p})=-1} \frac{1}{1-p^{-2s}} \prod_{(\frac{-1}{p})=1} \frac{1}{(1-p^{-s})^2} \right\}^2 \prod_p (1-p^{-2s}) \\
(2.4) \quad &= \frac{(4\zeta(s)L_{-4}(s))^2}{(1+2^{-s})\zeta(2s)}.
\end{aligned}$$

However, when $N \geq 3$, the generating functions cannot be expressed in terms of L -functions as completely as in formula (2.4). For example, when $N = 3$

$$\sum_{n=1}^{\infty} \frac{r_2^3(n)}{n^s} = \frac{64}{1-2^{-s}} \prod_{(\frac{-1}{p})=-1} \frac{1}{1-p^{-2s}} \prod_{(\frac{-1}{p})=1} \frac{1+4p^{-s}+p^{-2s}}{(1-p^{-s})^4},$$

and when $N = 4$

$$\sum_{n=1}^{\infty} \frac{r_2^4(n)}{n^s} = \frac{256}{1-2^{-s}} \prod_{(\frac{-1}{p})=-1} \frac{1}{1-p^{-2s}} \prod_{(\frac{-1}{p})=1} \frac{1+11p^{-s}+11p^{-2s}+p^{-3s}}{(1-p^{-s})^5}.$$

This helps explain why our Theorem 2.1 has no ‘closed-form’ extension to higher order. For the detailed asymptotic estimate of the generating function of the k th moment of $r_2(n)$, we refer the reader to [6].

3. SUMS OF A SMALL EVEN NUMBER OF SQUARES

In view of Theorem 2.1, whenever a Dirichlet series is expressible as a sum of two-fold products of L -functions:

$$L_f(s) = \sum_{\chi_1, \chi_2} a(\chi_1, \chi_2) L_{\chi_1}(s) L_{\chi_2}(s),$$

we are able to provide a closed form (in terms of L -functions) of the Dirichlet series $L_{f^2}(s) = \sum_{n=1}^{\infty} f^2(n)n^{-s}$, on using (2.1).

In particular, let $r_N(n)$ be the number of solutions to $x_1^2 + x_2^2 + \cdots + x_N^2 = n$ (counting permutations and signs) and let

$$L_N(s) := \sum_{n=1}^{\infty} r_N(n) n^{-s}, \quad R_N(s) := \sum_{n=1}^{\infty} r_N^2(n) n^{-s}$$

be the Dirichlet series corresponding to $r_N(n)$ and $r_N^2(n)$. Closed forms are obtainable, via the Mellin transform, for $L_N(s)$ for certain even N from the explicit formulae known for $r_N(n)$ (e.g. §91 in [15]). For example, we have

$$(3.1) \quad L_2(s) = 4\zeta(s)\beta(s),$$

$$(3.2) \quad L_4(s) = 8(1 - 4^{1-s})\zeta(s)\zeta(s-1),$$

$$(3.3) \quad L_6(s) = 16\zeta(s-2)\beta(s) - 4\zeta(s)\beta(s-2),$$

$$(3.4) \quad L_8(s) = 16(1 - 2^{1-s} + 4^{2-s})\zeta(s)\zeta(s-3),$$

where $\beta(s) := L_{-4}(s) = \sum_{n \geq 0} (-1)^n (2n+1)^{-s}$. In this section, we shall demonstrate how to use our Theorem 2.1 to obtain counterpart closed forms for $R_N(s)$ from the above expressions for $L_N(s)$.

Let us start with $R_2(s)$. It has already been shown in (2.4) that

$$R_2(s) = \sum_{n=1}^{\infty} \frac{r_2^2(n)}{n^s} = \frac{(4\zeta(s)L_{-4}(s))^2}{(1+2^{-s})\zeta(2s)}$$

but it can also be deduced directly from our Theorem 2.1 and (3.1) by taking $f_1(n) = f_2(n) = 1$ and $g_1(n) = g_2(n) = \left(\frac{-4}{n}\right)$.

We shall consider $R_4(s)$ and $R_8(s)$ later. For $R_6(s)$, we first write

$$\begin{aligned} L_6(s) &= 16\zeta(s-2)\beta(s) - 4\zeta(s)\beta(s-2) \\ &= 16 \sum_{n=1}^{\infty} \left(\sum_{d|n} d^2 \left(\frac{-4}{n/d} \right) \right) n^{-s} - 4 \sum_{n=1}^{\infty} \left(\sum_{d|n} d^2 \left(\frac{-4}{d} \right) \right) n^{-s} \\ &= \sum_{n=1}^{\infty} (16(f_1 * g_1)(n) - 4(f_2 * g_2)(n)) n^{-s} \end{aligned}$$

where $f_1(n) = n^2$, $g_1(n) = \left(\frac{-4}{n}\right)$, $f_2(n) = 1$ and $g_2(n) = \left(\frac{-4}{n}\right) n^2$. It follows from our Theorem 2.1 and (3.3) that

$$\begin{aligned} R_6(s) &= \sum_{n=1}^{\infty} (16(f_1 * g_1)(n) - 4(f_2 * g_2)(n))^2 n^{-s} \\ &= 16^2 \sum_{n=1}^{\infty} (f_1 * g_1)^2(n) n^{-s} - 128 \sum_{n=1}^{\infty} (f_1 * g_1)(n)(f_2 * g_2)(n) n^{-s} \\ &\quad + 16 \sum_{n=1}^{\infty} (f_2 * g_2)^2(n) n^{-s} \\ &= 16^2 \frac{L_{f_1^2}(s) L_{g_1^2}(s) L_{f_1 g_1}(s)^2}{L_{f_1^2 g_1^2}(2s)} - 128 \frac{L_{f_1 f_2}(s) L_{g_1 g_2}(s) L_{f_1 g_2}(s) L_{g_1 f_2}(s)}{L_{f_1 f_2 g_1 g_2}(2s)} \\ &\quad + 16 \frac{L_{f_2^2}(s) L_{g_2^2}(s) L_{f_2 g_2}(s)^2}{L_{f_2^2 g_2^2}(2s)}. \end{aligned} \tag{3.5}$$

It remains to evaluate the component L -functions and they are

$$\begin{aligned} L_{f_1^2}(s) &= \zeta(s-4), & L_{g_1^2}(s) &= (1-2^{-s})\zeta(s), \\ L_{f_2^2}(s) &= \zeta(s), & L_{g_2^2}(s) &= (1-16 \cdot 2^{-s})\zeta(s-4), \\ L_{f_1 g_1}(s) &= \beta(s-2), & L_{f_1 f_2}(s) &= \zeta(s-2), & L_{g_1 g_2}(s) &= (1-4 \cdot 2^{-s})\zeta(s-2), \\ L_{f_1 g_2}(s) &= \beta(s-4), & L_{g_1 f_2}(s) &= \beta(s), & L_{f_2 g_2}(s) &= \beta(s-2), \\ L_{f_1^2 g_1^2}(s) &= L_{f_2^2 g_2^2}(s) = L_{f_1 f_2 g_1 g_2}(s) &= (1-16 \cdot 2^{-s})\zeta(s-4). \end{aligned}$$

Now from (3.5), we have

$$\begin{aligned} R_6(s) &= 16 \frac{(17-32 \cdot 2^{-s}) \zeta(s-4) \beta^2(s-2) \zeta(s)}{(1-16 \cdot 2^{-2s}) \zeta(2s-4)} \\ &\quad - \frac{128}{(1+4 \cdot 2^{-s})} \frac{\beta(s-4) \zeta^2(s-2) \beta(s)}{\zeta(2s-4)}. \end{aligned}$$

For $R_4(s)$ and $R_8(s)$, we need the following companion lemma:

Lemma 3.1. *Suppose $f(n)$ is a multiplicative function. Let p be a prime and let the Dirichlet series*

$$\sum_{n=1}^{\infty} \frac{A(n)}{n^s} := \sum_{m=0}^{\infty} \frac{a_m}{p^{ms}} \sum_{n=1}^{\infty} \frac{f(n)}{n^s}$$

be the product of $L_f(s)$ and a power series in p^{-s} . Then

$$\begin{aligned} (3.6) \quad \sum_{n=1}^{\infty} \frac{A^2(n)}{n^s} &= L_{f^2}(s) \sum_{m=0}^{\infty} \frac{a_m^2}{p^{ms}} + 2L_{f^2}(s) \left(\sum_{l=0}^{\infty} \frac{f^2(p^l)}{p^{ls}} \right)^{-1} \\ &\quad \times \sum_{k=1}^{\infty} \left\{ \sum_{m=0}^{\infty} \frac{a_{m+k} a_m}{p^{ms}} \right\} \left\{ \sum_{l=0}^{\infty} \frac{f(p^l) f(p^{l+k})}{p^{ls}} \right\} p^{-ks}. \end{aligned}$$

Proof. Since

$$\sum_{n=1}^{\infty} A(n) n^{-s} = \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} a_m f(n) (p^m n)^{-s} = \sum_{n=1}^{\infty} \left\{ \sum_{\substack{m=0 \\ p^m | n}}^{\infty} a_m f\left(\frac{n}{p^m}\right) \right\} n^{-s},$$

we deduce

$$\begin{aligned} (3.7) \quad \sum_{n=1}^{\infty} A^2(n) n^{-s} &= \sum_{n=1}^{\infty} \left\{ \sum_{\substack{m=0 \\ p^m | n}}^{\infty} a_m f\left(\frac{n}{p^m}\right) \right\}^2 n^{-s} \\ &= \sum_{m_1, m_2=0}^{\infty} a_{m_1} a_{m_2} \sum_{\substack{n=1 \\ p^{m_1}, p^{m_2} | n}}^{\infty} f\left(\frac{n}{p^{m_1}}\right) f\left(\frac{n}{p^{m_2}}\right) n^{-s}. \end{aligned}$$

For any $m_1, m_2 \geq 1$ we let $M := \max(m_1, m_2)$ and $m := \min(m_1, m_2)$. Then the last summation (over n) in (3.7) is

$$\begin{aligned}
 &= \sum_{\substack{n=1 \\ p^M | n}}^{\infty} f\left(\frac{n}{p^M}\right) f\left(\frac{n}{p^m}\right) n^{-s} \\
 &= \frac{1}{p^{Ms}} \sum_{n=1}^{\infty} f(n) f(np^{M-m}) n^{-s} \\
 &= \frac{1}{p^{Ms}} \sum_{l=0}^{\infty} \sum_{\substack{n=1 \\ (p,n)=1}}^{\infty} f(np^l) f(np^{M-m+l}) p^{-ls} n^{-s} \\
 (3.8) \quad &= \frac{1}{p^{Ms}} \sum_{l=0}^{\infty} f(p^l) f(p^{M-m+l}) p^{-ls} \sum_{\substack{n=1 \\ (p,n)=1}}^{\infty} \frac{f^2(n)}{n^s}
 \end{aligned}$$

since $f(n)$ is multiplicative. By writing

$$\sum_{n=1}^{\infty} \frac{f^2(n)}{n^s} = \sum_{l=0}^{\infty} \sum_{\substack{n=1 \\ (p,n)=1}}^{\infty} \frac{f^2(np^l)}{(np^l)^s} = \sum_{l=0}^{\infty} \frac{f^2(p^l)}{p^{ls}} \sum_{\substack{n=1 \\ (p,n)=1}}^{\infty} \frac{f^2(n)}{n^s},$$

we deduce that

$$(3.9) \quad \sum_{\substack{n=1 \\ (p,n)=1}}^{\infty} \frac{f^2(n)}{n^s} = L_{f^2}(s) \left(\sum_{l=0}^{\infty} f^2(p^l) p^{-ls} \right)^{-1}.$$

Using (3.7), (3.8) and (3.9), we have

$$\begin{aligned}
 (3.10) \quad \sum_{n=1}^{\infty} A^2(n) n^{-s} &= L_{f^2}(s) \left(\sum_{l=0}^{\infty} f^2(p^l) p^{-ls} \right)^{-1} \times \\
 &\quad \times \sum_{m_1, m_2=0}^{\infty} \frac{a_{m_1} a_{m_2}}{p^{\max(m_1, m_2)s}} \sum_{l=0}^{\infty} \frac{f(p^l) f(p^{l+|m_1-m_2|})}{p^{ls}}.
 \end{aligned}$$

The contribution corresponding to $m_1 = m_2$ in the above double summation is

$$(3.11) \quad \sum_{m=0}^{\infty} \frac{a_m^2}{p^{ms}} \sum_{l=0}^{\infty} \frac{f^2(p^l)}{p^{ls}}$$

and the contribution corresponding to $m_1 \neq m_2$ is

$$\begin{aligned}
 &= 2 \sum_{m_2 < m_1}^{\infty} \frac{a_{m_1} a_{m_2}}{p^{m_1 s}} \sum_{l=0}^{\infty} \frac{f(p^l) f(p^{l+m_1-m_2})}{p^{ls}} \\
 &= 2 \sum_{m=0}^{\infty} \sum_{k=1}^{\infty} \frac{a_{m+k} a_m}{p^{(m+k)s}} \sum_{l=0}^{\infty} \frac{f(p^l) f(p^{l+k})}{p^{ls}} \\
 (3.12) \quad &= 2 \sum_{k=1}^{\infty} \left\{ \sum_{m=0}^{\infty} \frac{a_{m+k} a_m}{p^{ms}} \right\} \left\{ \sum_{l=0}^{\infty} \frac{f(p^l) f(p^{l+k})}{p^{ls}} \right\} \frac{1}{p^{ks}}.
 \end{aligned}$$

Now (3.6) follows from (3.10), (3.11) and (3.12). $\square$

On applying Lemma 3.1 to (3.2) and (3.4) and using (2.2), we have

$$R_4(s) = 64 \frac{(8 \cdot 2^{3-3s} - 10 \cdot 2^{2-2s} + 2^{1-s} + 1)\zeta(s-2)\zeta^2(s-1)\zeta(s)}{(1 + 2^{1-s})\zeta(2s-2)};$$

and

$$R_8(s) = 256 \frac{(32 \cdot 2^{6-2s} - 3 \cdot 2^{3-s} + 1)\zeta(s-6)\zeta^2(s-3)\zeta(s)}{(1 + 2^{3-s})\zeta(2s-6)}.$$

Therefore, we have completed the proof of the following Theorem.

Theorem 3.2. *We may write*

$$R_2(s) = \frac{(4\zeta(s)\beta(s))^2}{(1 + 2^{-s})\zeta(2s)}, \quad \Re(s) > 1;$$

$$R_4(s) = 64 \frac{(8 \cdot 2^{3-3s} - 10 \cdot 2^{2-2s} + 2^{1-s} + 1)\zeta(s-2)\zeta^2(s-1)\zeta(s)}{(1 + 2^{1-s})\zeta(2s-2)}, \quad \Re(s) > 3;$$

$$R_6(s) = 16 \frac{(17 - 32 \cdot 2^{-s})}{(1 - 16 \cdot 2^{-2s})} \frac{\zeta(s-4)\beta^2(s-2)\zeta(s)}{\zeta(2s-4)} - \frac{128}{(1 + 4 \cdot 2^{-s})} \frac{\beta(s-4)\zeta^2(s-2)\beta(s)}{\zeta(2s-4)}, \quad \Re(s) > 5;$$

and

$$R_8(s) = 256 \frac{(32 \cdot 2^{6-2s} - 3 \cdot 2^{3-s} + 1)\zeta(s-6)\zeta^2(s-3)\zeta(s)}{(1 + 2^{3-s})\zeta(2s-6)} \quad \Re(s) > 7.$$

Since $\epsilon\zeta(1+\epsilon) \rightarrow 1$ as $\epsilon \rightarrow 0$, the value of the $\lim_{\epsilon \rightarrow 0} \epsilon R_N(N-1+\epsilon)$ at its largest pole is, respectively:

$$\lim_{\epsilon \rightarrow 0} \epsilon R_4(3+\epsilon) = 96\zeta(3) = 3W_4$$

$$\lim_{\epsilon \rightarrow 0} \epsilon R_6(5+\epsilon) = 240\zeta(5) = 5W_6$$

and

$$\lim_{\epsilon \rightarrow 0} \epsilon R_8(7+\epsilon) = \frac{4480}{17}\zeta(7) = 7W_8.$$

Using the method in §5 below in conjunction with Theorem 3.2, we have

$$(3.13) \quad \sum_{n \leq x} r_2^2(n) = 4x \log x + 4\alpha x + O(x^{\frac{3}{4}+\epsilon})$$

where $\alpha := 2\gamma + \frac{8}{\pi}L'_{-4}(1) - \frac{12}{\pi^2}\zeta'(2) + \frac{1}{3}\log 2 - 1 = 2.0166216 \dots$ and

$$\sum_{n \leq x} r_N^2(n) = W_N x^{N-1} + O(x^{N-5/3+\epsilon})$$

for $N = 4, 6, 8$ with W_N given by (1.3). The best possible estimate would appear to be $O(x^{N-2+\epsilon})$ as will be discussed elsewhere.

4. CLOSED FORMS FOR DIRICHLET SERIES OF QUADRATIC FORMS

There is a rich parallel theory of L-functions over imaginary quadratic fields. In this vein, let $r_{2,P}(n)$ be the number of solutions to $x^2 + Py^2 = n$ (again counting sign and order). Denote

$$L_{2,P}(s) := \sum_{n=1}^{\infty} r_{2,P}(n) n^{-s}, \quad R_{2,P}(s) := \sum_{n=1}^{\infty} r_{2,P}(n)^2 n^{-s}.$$

It is known that when the quadratic form $x^2 + Py^2$ has disjoint discriminants (that is, it has exactly one form per genus), then one has the following formula (see (9.2.8) in [2])

$$(4.1) \quad \begin{aligned} L_{2,P} &= 2^{1-t} \sum_{\mu|P} L_{\epsilon_{\mu}\mu}(s) L_{-4P\epsilon_{\mu}/\mu}(s) \\ &= \sum_{n=1}^{\infty} \left\{ 2^{1-t} \sum_{\mu|P} \left(\frac{\epsilon_{\mu}\mu}{n} \right) * \left(\frac{-4P\epsilon_{\mu}/\mu}{n} \right) \right\} n^{-s} \end{aligned}$$

where P is an odd square-free number, t is the number of distinct factors of P and $\epsilon_{\mu} := \left(\frac{-1}{\mu} \right)$.

Explicitly, (4.1) holds for all *type one* numbers. These include and may comprise:

$$P = 5, 13, 21, 33, 37, 57, 85, 93, 105, 133, 165, 177, 253, 273, 345, 357, 385, 1365.$$

It is known that there are only finitely many such disjoint discriminants. We call such P **solvable**. Using (4.1), we have

$$\begin{aligned} R_{2,P}(s) &= \sum_{n=1}^{\infty} 2^{2-2t} \sum_{\mu_1\mu_2|P} \left[\left(\frac{\epsilon_{\mu_1}\mu_1}{n} \right) * \left(\frac{-4P\epsilon_{\mu_1}/\mu_1}{n} \right) \right] \cdot \left[\left(\frac{\epsilon_{\mu_2}\mu_2}{n} \right) * \left(\frac{-4P\epsilon_{\mu_2}/\mu_2}{n} \right) \right] n^{-s} \\ &= 2^{2-2t} \sum_{\mu_1\mu_2|P} \sum_{n=1}^{\infty} \left[\left(\frac{\epsilon_{\mu_1}\mu_1}{n} \right) * \left(\frac{-4P\epsilon_{\mu_1}/\mu_1}{n} \right) \right] \cdot \left[\left(\frac{\epsilon_{\mu_2}\mu_2}{n} \right) * \left(\frac{-4P\epsilon_{\mu_2}/\mu_2}{n} \right) \right] n^{-s}. \end{aligned}$$

We now notice that $R_{2,P}(s)$ is a sum of Dirichlet series in the form of Theorem 2.1. We may apply Theorem 2.1 on letting

$$f_i(n) := \left(\frac{\epsilon_{\mu_i}\mu_i}{n} \right), \quad g_i(n) := \left(\frac{-4P\epsilon_{\mu_i}/\mu_i}{n} \right),$$

for $i = 1, 2$. Then

$$\begin{aligned} L_{f_1 f_2}(s) &= \sum_{n=1}^{\infty} \left(\frac{\epsilon_{\mu_1}\mu_1}{n} \right) \left(\frac{\epsilon_{\mu_2}\mu_2}{n} \right) n^{-s} \\ &= \sum_{\substack{n=1 \\ (n, (\mu_1, \mu_2))=1}}^{\infty} \left(\frac{\epsilon_{\mu_1^*}\mu_2^*\mu_1^*\mu_2^*}{n} \right) n^{-s} \\ &= L_{\epsilon_{\mu_1^*}\mu_2^*\mu_1^*\mu_2^*}(s) \prod_{p|(\mu_1, \mu_2)} \left(1 - \left(\frac{\epsilon_{\mu_1^*}\mu_2^*\mu_1^*\mu_2^*}{p} \right) p^{-s} \right) \end{aligned}$$

where $\mu_i^* := \mu_i/(\mu_1, \mu_2)$ and $\prod_{p|n}$ denotes the product over all prime factors of n . Similarly, we have

$$\begin{aligned} L_{g_1 g_2}(s) &= L_{\epsilon_{\mu_1^* \mu_2^*} \mu_1^* \mu_2^*}(s) \prod_{p|\frac{2P}{[\mu_1, \mu_2]}} \left(1 - \left(\frac{\epsilon_{\mu_1^* \mu_2^*} \mu_1^* \mu_2^*}{p}\right) p^{-s}\right); \\ L_{f_1 g_2}(s) &= L_{-4P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}(s) \prod_{p|\mu_1^*} \left(1 - \left(\frac{-4P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}{p}\right) p^{-s}\right); \\ L_{f_2 g_1}(s) &= L_{-4P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}(s) \prod_{p|\mu_2^*} \left(1 - \left(\frac{-4P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}{p}\right) p^{-s}\right) \end{aligned}$$

and

$$L_{f_1 f_2 g_1 g_2}(s) = \zeta(s) \prod_{p|2P} (1 - p^{-s}).$$

Our basic Theorem 2.1 gives

$$\begin{aligned} R_{2,P}(s) &= 2^{2(1-t)} \sum_{\mu_1, \mu_2 | P} L_{\epsilon_{\mu_1^* \mu_2^*} \mu_1^* \mu_2^*}^2(s) L_{-4P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}^2(s) \zeta(2s)^{-1} \\ &\quad \times \prod_{p|2P} \left\{ 1 + \left[\left(\frac{\epsilon_{\mu_1^* \mu_2^*} \mu_1^* \mu_2^*}{p} \right) + \left(\frac{-4P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}{p} \right) \right] p^{-s} \right\}^{-1}. \end{aligned}$$

We have similar closed forms of L -functions for the quadratic form $x^2 + 2Py^2$ with discriminant $-8P$ (see (9.2.9) in [2]):

$$L_{2,2P} = 2^{1-t} \sum_{\mu|P} L_{\epsilon_\mu \mu}(s) L_{-8P\epsilon_\mu/\mu}(s).$$

We deduce from Theorem 2.1, in the same way, that

$$\begin{aligned} R_{2,2P}(s) &= 2^{2(1-t)} \sum_{\mu_1, \mu_2 | P} L_{\epsilon_{\mu_1^* \mu_2^*} \mu_1^* \mu_2^*}^2(s) L_{-8P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}^2(s) \zeta(2s)^{-1} \\ &\quad \times \prod_{p|2P} \left\{ 1 + \left[\left(\frac{\epsilon_{\mu_1^* \mu_2^*} \mu_1^* \mu_2^*}{p} \right) + \left(\frac{-8P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}{p} \right) \right] p^{-s} \right\}^{-1} \end{aligned}$$

for the *type two* integers

$$P = 1, 3, 5, 11, 15, 21, 29, 35, 39, 51, 65, 95, 105, 165, 231.$$

We note that $210 = 2 \times 105$ yields the invariant which Ramanujan sent to Hardy in his famous letter.

We may reprise with the following theorem:

Theorem 4.1. *Let P be a solvable square-free integer and let t be the number of distinct factors of P . We have for P respectively of type one and type two:*

$$\begin{aligned} (4.2) \quad R_{2,P}(s) &= 2^{2(1-t)} \sum_{\mu_1, \mu_2 | P} L_{\epsilon_{\mu_1^* \mu_2^*} \mu_1^* \mu_2^*}^2(s) L_{-4P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}^2(s) \zeta(2s)^{-1} \\ &\quad \times \prod_{p|2P} \left\{ 1 + \left[\left(\frac{\epsilon_{\mu_1^* \mu_2^*} \mu_1^* \mu_2^*}{p} \right) + \left(\frac{-4P\epsilon_{\mu_1^* \mu_2^*}/\mu_1^* \mu_2^*}{p} \right) \right] p^{-s} \right\}^{-1}, \end{aligned}$$

and

$$R_{2,2P}(s) = 2^{2(1-t)} \sum_{\mu_1, \mu_2 | P} L_{\epsilon_{\mu_1}^* \mu_2^* \mu_1^* \mu_2^*}(s) L_{-8P \epsilon_{\mu_1}^* \mu_2^* / \mu_1^* \mu_2^*}(s) \zeta(2s)^{-1} \\ \times \prod_{p|2P} \left\{ 1 + \left[\left(\frac{\epsilon_{\mu_1}^* \mu_2^* \mu_1^* \mu_2^*}{p} \right) + \left(\frac{-8P \epsilon_{\mu_1}^* \mu_2^* / \mu_1^* \mu_2^*}{p} \right) \right] p^{-s} \right\}^{-1}$$

where $\epsilon_\mu = \left(\frac{-1}{\mu} \right)$ and $\mu_i^* = \mu_i / (\mu_1, \mu_2)$.

In particular, the prime cases provide:

Corollary 4.2. *We have*

$$R_{2,p}(s) = \frac{2\zeta^2(s) L_{-4p}^2(s)}{(1+2^{-s})(1+p^{-s})\zeta(2s)} + \frac{2L_p^2(s) L_{-4}^2(s)}{(1-2^{-s})(1+p^{-s})\zeta(2s)}$$

for $p = 5, 13, 37$, while

$$R_{2,2}(s) = \frac{4\zeta^2(s) L_{-8}^2(s)}{(1+2^{-s})\zeta(2s)}.$$

Similarly,

$$R_{2,2p}(s) = \frac{2\zeta^2(s) L_{-8p}^2(s)}{(1+2^{-s})(1+p^{-s})\zeta(2s)} + \frac{2L_{-p}^2(s) L_8^2(s)}{(1-2^{-s})(1-p^{-s})\zeta(2s)},$$

for $p = 3, 11$ while

$$R_{2,2p}(s) = \frac{2\zeta^2(s) L_{-8p}^2(s)}{(1+2^{-s})(1+p^{-s})\zeta(2s)} + \frac{2L_p^2(s) L_{-8}^2(s)}{(1-2^{-s})(1-p^{-s})\zeta(2s)}$$

for $p = 5, 29$.

Closed forms for $L_{2,P}(s)$ are also accessible for some P other than those of *type one* or *type two*. For example, (see Table VI of [9]) one has

$$(4.3) \quad L_{2,3}(s) = (2 + 4^{1-s}) \zeta(s) L_{-3}(s).$$

and hence by Theorem 2.1 and Lemma 3.1, we obtain

$$(4.4) \quad R_{2,3}(s) = 4 \frac{1 + 2^{3-2s}}{1 + 3^{-s}} \frac{(\zeta(s) L_{-3}(s))^2}{\zeta(2s)}.$$

We may also derive many formulae for non-square free integers via modular transformations [2]. We contain ourselves with the simplest example which is

$$R_{2,4}(s) = \frac{4 - 2^{2-s} + 2^{4-2s}}{1 + 2^{-s}} \frac{(\zeta(s) \beta(s))^2}{\zeta(2s)}$$

as a consequence of a quadratic transformation leading to

$$L_{2,4}(s) = (2^{-1} - 2^{-1-s} + 4^{-s}) L_2(s).$$

There are some simple closed forms of the generating functions for more general binary quadratic forms found in [9]. Let

$$L_{(a,b,c)}(s) := \sum_{(n,m) \neq (0,0)} \frac{1}{(am^2 + bmn + cn^2)^s} = \sum_{n=1}^{\infty} \frac{r_{(a,b,c)}(n)}{n^s}$$

and $R_{(a,b,c)}(s) := \sum_{n=1}^{\infty} \frac{r_{(a,b,c)}(n)^2}{n^s}$ where $r_{(a,b,c)}(n)$ is the number of representations of n by the quadratic form $ax^2 + bxy + cy^2$. Then, we have (e.g. (26) of [18])

$$\sum_{h(D)} L_{(a,b,c)}(s) = \omega(D)\zeta(s)L_D(s)$$

where the sum is taken over the $h(D)$ inequivalent reduced quadratic forms of discriminant $D := b^2 - 4ac$ and $\omega(-3) = 6$, $\omega(-4) = 4$ and $\omega(D) = 2$ for $D < -4$. In particular, for $c = 2, 3, 5, 11, 17, 41$, $h(D) = 1$ and the result is especially simple:

$$L_{(1,1,c)}(s) = 2\zeta(s)L_D(s).$$

Hence from Theorem 2.1, we have

$$R_{(1,1,c)}(s) = \frac{4(\zeta(s)L_D(s))^2}{(1 + |D|^{-s})\zeta(2s)},$$

with similar formulae for $(a, b, c) = (1, 1, 1)$ and $(1, 0, 1)$.

Thanks to the *On-Line Encyclopedia of Integer Sequences*

<http://www.research.att.com/~njas/sequences/>

we discover that the sequence 2, 3, 5, 11, 17, 41 is exactly the so-called Euler “lucky” numbers which are the numbers n such that $m \rightarrow m^2 - m + n$ has prime values for $m = 0, \dots, n - 1$.

5. THE AVERAGE ORDER OF $r_{2,P}(n)$

We start with the average order of $r_{2,P}$. Our method would seem to add an unnecessary if unobtrusive ‘ ϵ ’.

Theorem 5.1. *Let P be a solvable square-free integer, $x > 1$ and $\epsilon > 0$. We have for either $N = P$ of type one or $N = 2P$ of type two:*

$$\sum_{n \leq x} r_{2,N}(n) = \frac{\pi}{\sqrt{N}}x + O((xN)^{\frac{1}{2}+\epsilon}).$$

where the implicit constants are independent of x and P .

Proof. In view of (4.1), we have for $n \geq 1$

$$(5.1) \quad r_{2,P}(n) = 2^{1-t} \sum_{\mu|P} \left(\frac{\epsilon_{\mu}\mu}{n} \right) * \left(\frac{-4P\epsilon_{\mu}/\mu}{n} \right) \leq 2^{1-t} \sum_{\mu|P} \sigma_0(n) \leq 2\sigma_0(n).$$

It follows from (1.1) that

$$L_{2,P}(\sigma) \ll \sum_{n=1}^{\infty} \frac{\sigma_0(n)}{n^{\sigma}} = \zeta(\sigma)^2 \ll \frac{1}{(\sigma-1)^2}$$

as $\sigma \rightarrow 1^+$. Now in view of Perron’s formula (see Theorem 1 in §1 of Chapter V in [14]), for any $c > 1$, $\epsilon > 0$ and $x, T \geq 1$ we have

$$(5.2) \quad \sum_{n \leq x} r_{2,P}(n) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} L_{2,P}(s) \frac{x^s}{s} ds + O(x^c T^{-1}(c-1)^{-2} + x^{1+\epsilon} T^{-1}).$$

In order to evaluate the above integral, we need the following well-known estimates for $\zeta(s)$ and L -functions.

Lemma 5.2. *We have*

$$\zeta(\sigma + i\xi) \ll \begin{cases} \frac{1}{\sigma-1} & \text{if } 1 < \sigma \leq 2 \text{ and } \xi = 0 \\ \log |\xi| & \text{if } 1 \leq \sigma \text{ and } |\xi| \geq e \\ |\xi|^{\frac{1-\sigma}{2}} \log |\xi| & \text{if } 0 \leq \sigma \leq 1 \text{ and } |\xi| \geq e \end{cases}$$

and

$$\frac{1}{\zeta(\sigma + i\xi)} \ll \log^7 |\xi|$$

if $\sigma \geq 1$ and $|\xi| \geq e$. If χ is a non-principal character modulo q , we have

$$L(\sigma + i\xi, \chi) \ll \log q(|\xi| + 2)$$

for $\sigma \geq 1$ while if χ is a primitive character modulo $q \geq 3$ and $0 \leq \sigma \leq 1$, then

$$L(\sigma + i\xi, \chi) \ll (q(|\xi| + 2))^{\frac{1-\sigma}{2}} \log q(|\xi| + 2).$$

As usual, we estimate the integral in (5.2) by replacing the integral over the rectangle R with vertices $b \pm iT$ and $c \pm iT$ with $b = \frac{1}{\log x}$ and then calculate the residues of the poles of the integrand inside R . In view of (4.2), the only pole of $R_{2,P}(s) \frac{x^s}{s}$ inside R is $s = 1$, which comes from $\zeta(s)$, and its residue at $s = 1$ is $2^{1-t} L_{-4P}(1)x$ because $\lim_{s \rightarrow 1} (s-1)\zeta(s) = 1$.

For solvable P , i.e. $x^2 + Py^2$ having one form per genus, the class number equals the number of genera — which is 2^t (see p. 198 of [17]). Hence $L_{-4P}(1) = \frac{2^{t-1}\pi}{\sqrt{P}}$ for type one P and $L_{-8P}(1) = \frac{2^{t-1}\pi}{\sqrt{2P}}$ for type two P by (4.11) in [10]. Thus, the residue of $R_{2,P}(s) \frac{x^s}{s}$ at $s = 1$ is $\frac{\pi}{\sqrt{P}}x$.

Next, using the estimates in Lemma 5.2 and (4.2), we may prove that for $|\xi| \leq T$,

$$L_{2,P}(\sigma + i\xi) \ll \begin{cases} (P(|\xi| + 2))^{(1-\sigma)} \log^2(PT) & \text{if } b \leq \sigma \leq 1, \\ \log^2(PT) & \text{if } 1 \leq \sigma \leq c. \end{cases}$$

It then follows that

$$\begin{aligned} \frac{1}{2\pi i} \int_{b-iT}^{b+iT} L_{2,P}(s) \frac{x^s}{s} ds &\ll \int_{-T}^T |L_{2,P}(b + i\xi)| \frac{x^b}{|b + i\xi|} d\xi \\ (5.3) \quad &\ll PT \log^2(PT) \end{aligned}$$

and

$$\begin{aligned} &\frac{1}{2\pi i} \int_{b \pm iT}^{c \pm iT} L_{2,P}(s) \frac{x^s}{s} ds \\ &\ll \left\{ \int_b^1 + \int_1^c \right\} |L_{2,P}(\sigma \pm iT)| \frac{x^\sigma}{T} d\sigma \\ &\ll P(\log PT)^2 \int_b^1 \left(\frac{x}{PT} \right)^\sigma d\sigma + T^{-1} (\log PT)^2 \int_1^c x^\sigma d\sigma \\ (5.4) \quad &\ll x^c T^{-1} \log^2(PT) \log x. \end{aligned}$$

Now by choosing $c = 1 + \frac{1}{\log x}$ and $T = (x/P)^{\frac{1}{2}}$, we get from (5.2)–(5.4) that

$$\sum_{n \leq x} r_{2,P}(n) = \frac{\pi}{\sqrt{P}}x + O((xP)^{\frac{1}{2}+\epsilon}).$$

The case for type two P can be proved in the same way. This completes the proof of Theorem 5.1. $\square$

For any square-free integer N , we define a constant α by:

$$(5.5) \quad \alpha(N) := 2\gamma + \sum_{p|2N} \frac{\log p}{p+1} + 2 \frac{L'_{-4N}(1)}{L_{-4N}(1)} - \frac{12}{\pi^2} \zeta'(2) - 1$$

where γ is Euler's constant and $\sum_{p|n}$ is the summation over all prime factors of n .

Theorem 5.3. *Let P be a solvable square-free integer. Let $x > 1$ and $\epsilon > 0$. We have for either $N = P$ of type one or $N = 2P$ of type two:*

$$\sum_{n \leq x} r_{2,N}(n)^2 = \frac{3}{N} \left(\prod_{p|2N} \frac{2p}{p+1} \right) (x \log x + \alpha(N)x) + O(N^{\frac{1}{4}+\epsilon} x^{\frac{3}{4}+\epsilon})$$

where the implicit constants are independent of both x and P .

Proof. It follows from (1.2) and (5.1) that

$$R_{2,P}(\sigma) \ll \sum_{n=1}^{\infty} \frac{\sigma_0(n)^2}{n^\sigma} = \frac{\zeta^4(\sigma)}{\zeta(2\sigma)} \ll \frac{1}{(\sigma-1)^4}$$

as $\sigma \rightarrow 1^+$. Similar to (5.2), for any $c > 1$, $\epsilon > 0$ and $x, T \geq 1$, we have

$$(5.6) \quad \sum_{n \leq x} r_{2,P}(n)^2 = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} R_{2,P}(s) \frac{x^s}{s} ds + O(x^c T^{-1} (c-1)^{-4} + x^{1+\epsilon} T^{-1}).$$

We estimate the integral in (5.3) by replacing the integral over the rectangle R with vertices $\frac{1}{2} \pm iT$ and $c \pm iT$ and then calculate the residues of the poles of the integrand inside R . In view of (4.2), the only pole of $R_{2,P}(s) \frac{x^s}{s}$ inside R is $s = 1$ of order 2 which comes from $\zeta(s)^2$ and corresponds to the terms when $\mu_1 = \mu_2$ in the double summation of (4.2):

$$(5.7) \quad 2^{2(1-t)} \sigma_0(P) \zeta(s)^2 L_{-4P}(s)^2 \zeta(2s)^{-1} \prod_{p|2P} (1+p^{-s})^{-1} \frac{x^s}{s} := F(s)$$

and its residue at $s = 1$ is

$$\begin{aligned} &= \lim_{s \rightarrow 1} \frac{d}{ds} \{ (s-1)^2 F(s) \} \\ &= \lim_{s \rightarrow 1} (s-1)^2 F(s) \lim_{s \rightarrow 1} \frac{d}{ds} \log \{ (s-1)^2 F(s) \}. \end{aligned}$$

Since P is solvable, so

$$\begin{aligned} \lim_{s \rightarrow 1} (s-1)^2 F(s) &= 2^{2(1-t)} \sigma_0(P) L_{-4P}^2(1) \zeta(2)^{-1} \prod_{p|2P} (1+p^{-1})^{-1} x \\ &= \frac{3}{P} \left(\prod_{p|2P} \frac{2p}{p+1} \right) x. \end{aligned}$$

In view of (5.5) and (5.7), we have

$$\begin{aligned}
& \lim_{s \rightarrow 1} \frac{d}{ds} \log \{(s-1)^2 F(s)\} \\
&= 2\gamma + \sum_{p|2P} \frac{\log p}{p+1} + 2 \frac{L'_{-4P}(1)}{L_{-4P}(1)} - \frac{12}{\pi^2} \zeta'(2) - 1 + \log x \\
&= \alpha(P) + \log x
\end{aligned}$$

because $\lim_{s \rightarrow 1} \left(\frac{1}{s-1} + \frac{\zeta'(s)}{\zeta(s)} \right) = \gamma$. Therefore the residue of $R_{2,P}(s) \frac{x^s}{s}$ at $s = 1$ is

$$(5.8) \quad \frac{3}{P} \left(\prod_{p|2P} \frac{2p}{p+1} \right) (x \log x + \alpha(P)x).$$

Next using the estimates in Lemma 5.2 and (4.2), one can prove that for $|\xi| \leq T$,

$$R_{2,P}(\sigma + i\xi) \ll \begin{cases} P^{(1-\sigma)+\epsilon} (|\xi| + 2)^{2(1-\sigma)} \log^A T & \text{if } \frac{1}{2} \leq \sigma \leq 1, \\ P^\epsilon \log^A T & \text{if } 1 \leq \sigma \leq c. \end{cases}$$

It then follows that

$$(5.9) \quad \begin{aligned} \frac{1}{2\pi i} \int_{\frac{1}{2}-iT}^{\frac{1}{2}+iT} R_{2,P}(s) \frac{x^s}{s} ds &\ll \int_{-T}^T |R_{2,P}(\frac{1}{2} + i\xi)| \frac{x^{\frac{1}{2}}}{|\frac{1}{2} + i\xi|} d\xi \\ &\ll P^{\frac{1}{2}+\epsilon} x^{\frac{1}{2}} T \log^A T \end{aligned}$$

and

$$\begin{aligned}
& \frac{1}{2\pi i} \int_{\frac{1}{2} \pm iT}^{c \pm iT} R_{2,P}(s) \frac{x^s}{s} ds \\
&\ll \left\{ \int_{\frac{1}{2}}^1 + \int_1^c \right\} |R_{2,P}(\sigma \pm iT)| \frac{x^\sigma}{T} d\sigma \\
&\ll P^{1+\epsilon} T (\log T)^A \int_{\frac{1}{2}}^1 \left(\frac{x}{PT^2} \right)^\sigma d\sigma + P^\epsilon T^{-1} (\log T)^A \int_1^c x^\sigma d\sigma \\
(5.10) \quad &\ll P^\epsilon x^c T^{-1} \log^A T.
\end{aligned}$$

Now by choosing $c = 1 + \frac{1}{\log x}$ and $T = (x/P)^{\frac{1}{4}}$, we get from (5.6) and (5.8)-(5.10) that

$$\sum_{n \leq x} r_{2,P}(n)^2 = \frac{3}{P} \left(\prod_{p|2P} \frac{2p}{p+1} \right) (x \log x + \alpha(P)x) + O(P^{\frac{1}{4}+\epsilon} x^{\frac{3}{4}+\epsilon}).$$

The case for type two P can be proved in the same way. This completes the proof of Theorem 5.3. $\square$

In particular, we have established:

Theorem 5.4. *For any $x \geq 1$, we have*

$$\sum_{n \leq x} r_{2,p}(n)^2 = \frac{8}{p+1} (x \log x + \alpha(p)x) + O(x^{\frac{3}{4}+\epsilon})$$

for $p = 5, 13, 37$ and

$$\sum_{n \leq x} r_{2,2p}(n)^2 = \frac{4}{p+1}(x \log x + \alpha(2p)x) + O(x^{\frac{3}{4}+\epsilon})$$

for $p = 1, 3, 5, 11, 29$. Here the implicit constants are again independent of x .

Similarly, in view of (4.3) and (4.4), we have for $x > 1$,

$$\sum_{n \leq x} r_{2,3}(n) = \frac{\pi}{\sqrt{3}}x + O(x^{\frac{1}{2}+\epsilon})$$

and

$$(5.11) \quad \sum_{n \leq x} r_{2,3}(n)^2 = 2(x \log x + \alpha_3 x) + O(x^{\frac{3}{4}+\epsilon})$$

where $\alpha_3 := 2\gamma - \frac{4}{3} \log 2 + \frac{1}{4} \log 3 + \frac{6\sqrt{3}}{\pi} L'_{-3}(1) - \frac{12}{\pi^2} \zeta'(2) - 1$.

Also

$$\sum_{n \leq x} r_{2,4}(n) = \frac{\pi}{2}x + O(x^{\frac{1}{2}+\epsilon})$$

and

$$\sum_{n \leq x} r_{2,4}(n)^2 = \frac{3}{2}(x \log x + \alpha_4 x) + O(x^{\frac{3}{4}+\epsilon})$$

where $\alpha_4 := 2\gamma - \frac{2}{3} \log 2 + \frac{8}{\pi} L'_{-4}(1) - \frac{12}{\pi^2} \zeta'(2) - 1$.

Akin to Wagon's conjecture, we make the following conjecture.

Quadratic Conjecture. *For any square-free P ,*

$$\sum_{n \leq x} r_{2,P}(n) \sim \frac{\pi}{\sqrt{P}}x$$

and

$$\sum_{n \leq x} r_{2,P}(n)^2 \sim \frac{3}{P} \left(\prod_{p|2P} \frac{2p}{p+1} \right) x \log x$$

as $x \rightarrow \infty$.

In view of Theorem 5.3, (3.13) and (5.11), our conjecture is true for solvable P and for $P = 1, 3$. We have also confirmed it for $P = 7$ and 15 from the representations of

$$L_{2,7}(s) = 2(1 - 2^{1-s} + 2^{1-2s})\zeta(s)L_{-7}(s)$$

and

$$L_{2,15}(s) = (1 - 2^{1-s} + 2^{1-2s})\zeta(s)L_{15}(s) + (1 + 2^{1-s} + 2^{1-2s})L_{-3}(s)L_5(s)$$

again given in [9], which leads to

$$R_{2,7}(s) = 4 \frac{(1 - 3 \cdot 2^{-s} + 2^{2-2s})}{(1 + 2^{-s})(1 + 7^{-s})} \frac{(\zeta(s)L_{-7}(s))^2}{\zeta(2s)}$$

and

$$\begin{aligned} R_{2,15}(s) &= \frac{2(1 - 3 \cdot 2^{-s} + 2^{2-2s})}{(1 + 2^{-s})(1 + 3^{-s})(1 + 5^{-s})} \frac{(\zeta(s)L_{-15}(s))^2}{\zeta(2s)} \\ &\quad + \frac{2(1 + 3 \cdot 2^{-s} + 2^{2-2s})}{(1 - 2^{-s})(1 - 3^{-s})(1 - 5^{-s})} \frac{(L_{-3}(s)L_5(s))^2}{\zeta(2s)}, \end{aligned}$$

and may be analyzed by the methods above.

6. SUMS OF THREE SQUARES AND OTHER POWERS

6.1. Three Squares. Odd squares are notoriously less amenable to closed forms. In this subsection, we primarily record some results for $r_3(n)$, the number of representations of n as a sum of three squares. Following Hardy and Bateman, Hua in [13] gives the following formula for $r_3(n)$. Let

$$\chi_2(n) := \begin{cases} 0 & \text{if } 4^{-a}n \equiv 7 \pmod{8}; \\ 2^{-a} & \text{if } 4^{-a}n \equiv 3 \pmod{8}; \\ 3 \cdot 2^{-1-a} & \text{if } 4^{-a}n \equiv 1, 2, 5, 6 \pmod{8} \end{cases}$$

where a is the highest power of 4 dividing n .

Then

$$(6.1) \quad r_3(n) = \frac{16\sqrt{n}}{\pi} L_{-4n}(1) \chi_2(n) \times \prod_{p^2|n} \left(\frac{p^{-\tau} - 1}{p^{-1} - 1} + p^{-\tau} \left(1 - \frac{1}{p} \left(\frac{-p^{-2\tau}n}{p} \right) \right)^{-1} \right)$$

where $\tau = \tau_p$ is the highest power of p^2 dividing n .

The Dirichlet series for $r_3(n)$ deriving from (6.1) is not as malleable as those of (3.1)-(3.4), but we are able to derive a nice expression in terms of Bessel functions.

Let K_s be the *modified Bessel function of the second kind*. Then we have (see [19], p. 183)

$$(6.2) \quad K_s(x) = \frac{1}{2} \left(\frac{x}{2} \right)^s \int_0^\infty e^{-t - \frac{x^2}{4t}} \frac{dt}{t^{s+1}}.$$

By the substitution $t = \frac{1}{u}$ in (6.2), we get

$$(6.3) \quad K_s(x) = \frac{1}{2} \left(\frac{x}{2} \right)^s \int_0^\infty e^{-\frac{x^2 u}{4} - \frac{1}{u}} u^{s-1} du.$$

Let

$$\theta_3(q) := \sum_{n=-\infty}^{\infty} q^{n^2}$$

be the classical Jacobean theta function. In view of the Poisson summation formula, we have, for $t > 0$

$$\theta_3(e^{-\pi t}) = t^{-\frac{1}{2}} \theta_3(e^{-\pi/t}).$$

Since the Mellin transform of $e^{-\alpha t}$ for $\alpha \neq 0$ is $M_s(e^{-\alpha t}) = \Gamma(s)\alpha^{-s}$, so we have (letting $q = e^{-\pi t}$)

$$\begin{aligned}
L_3(s) &= 3 \sum_{n,m,p \in \mathbb{Z}} \frac{n^2}{(n^2 + m^2 + p^2)^{s+1}} \\
&= \frac{3\pi^{s+1}}{\Gamma(s+1)} \sum_{n,m,p \in \mathbb{Z}} n^2 M_{s+1}(q^{n^2+m^2+p^2}) \\
&= \frac{3\pi^{s+1}}{\Gamma(s+1)} M_{s+1} \left(\sum_{n \in \mathbb{Z}} n^2 q^{n^2} \theta_3^2(q) \right) \\
&= \frac{3\pi^{s+1}}{\Gamma(s+1)} \sum_{n \in \mathbb{Z}} n^2 \int_0^\infty e^{-n^2 \pi t} \theta_3^2(e^{-\pi/t}) t^{s-1} dt \\
&= \frac{3\pi^{s+1}}{\Gamma(s+1)} \sum_{n \in \mathbb{Z}} n^2 \sum_{m=1}^\infty r_2(m) \int_0^\infty e^{-n^2 \pi t - \frac{\pi m}{t}} t^{s-1} dt \\
&\quad + \frac{3\pi^{s+1}}{\Gamma(s+1)} \sum_{n \in \mathbb{Z}} n^2 \int_0^\infty e^{-n^2 \pi t} t^{s-1} dt.
\end{aligned} \tag{6.4}$$

The first term of (6.4) is

$$\begin{aligned}
&= \frac{6\pi^{s+1}}{\Gamma(s+1)} \sum_{n=1}^\infty n^2 \sum_{m=1}^\infty r_2(m) \int_0^\infty e^{-n^2 \pi t - \frac{\pi m}{t}} t^{s-1} dt \\
&= \frac{6\pi^{s+1}}{\Gamma(s+1)} \sum_{m=1}^\infty r_2(m) (\pi m)^s \sum_{n=1}^\infty n^2 \int_0^\infty e^{-n^2 \pi^2 m x - 1/x} x^{s-1} dx, \quad (x = \frac{t}{\pi m}) \\
&= \frac{12\pi^{s+1}}{\Gamma(s+1)} \sum_{m=1}^\infty r_2(m) m^{s/2} \sum_{n=1}^\infty \frac{1}{n^{s-2}} K_s(2\pi n \sqrt{m})
\end{aligned}$$

by (6.3) and the second term is

$$\begin{aligned}
&= \frac{6\pi^{s+1}}{\Gamma(s+1)} \sum_{n=1}^\infty \frac{1}{n^{2s-2} \pi^s} \int_0^\infty e^{-x} x^{s-1} dx \\
&= \frac{6\pi}{s} \zeta(2s-2).
\end{aligned}$$

This proves the following result:

$$(6.5) \quad L_3(s) = \frac{6\pi}{s} \zeta(2s-2) + \frac{12\pi^{s+1}}{\Gamma(s+1)} \sum_{m=1}^\infty r_2(m) m^{s/2} \sum_{n=1}^\infty \frac{1}{n^{s-2}} K_s(2\pi n \sqrt{m}).$$

There is a corresponding formula for $\sum (-1)^n r_3(n)/n^s$ which corresponds to Madelung's constant (see p. 301 in [2]). The second term of (6.5) can be rewritten as

$$\frac{12\pi^{s+1}}{\Gamma(s+1)} \sum_{k>0} k^{\frac{s}{2}} K_s(2\pi \sqrt{k}) \sum_{n^2|k} \frac{r_2(k/n^2)}{n^{2s-2}}.$$

Moreover, these Bessel functions are elementary when s is a half-integer. Most nicely, for 'jellium', which is the Wigner sum analogue of Madelung's constant, we

have

$$L_3(1/2) = -\pi + 3\pi \sum_{m>0} \frac{r_2(m)}{\sinh^2(\pi\sqrt{m})},$$

and the exponential convergence is entirely apparent.

For a survey of other rapidly convergent lattice sums of this type see [2] and [4].

There is a corresponding formula for $L_N(s)$, for all $N \geq 2$, in which we obtain a Bessel-series in $r_{N-1}(m)$:

$$(6.6) \quad \begin{aligned} L_N(s) = \sum_{n>0} \frac{r_N(n)}{n^s} &= \frac{2N \Gamma(s - \frac{N-3}{2})}{\Gamma(s+1)} \pi^{\frac{N-1}{2}} \zeta(2s - N + 1) \\ &+ \frac{4N \pi^{s+1}}{\Gamma(s+1)} \sum_{m>0} \frac{m^{\frac{1}{2}s} r_{N-1}(m)}{m^{\frac{N-3}{4}}} \sum_{n>0} \frac{n^{\frac{N+1}{2}}}{n^s} K_{s - \frac{N-3}{2}}(2n\pi\sqrt{m}). \end{aligned}$$

There is an equally attractive integral representation (see [19] p. 172) for:

$$K_s(x) = \left(\frac{2}{x}\right)^s \frac{\Gamma(s+1/2)}{\Gamma(1/2)} \int_0^\infty \frac{\cos(xt)}{(1+t^2)^{s+1/2}} dt$$

at least when $x > 1/2$. This leads to

$$\sum_{n>0} \frac{r_3(n)}{n^s} = 2\beta(s + \frac{1}{2}, \frac{1}{2}) \sum_{m>0} r_2(m) \int_0^\infty \frac{C_{s-2}(\sqrt{mt})}{(1+t^2)^{s+1/2}} dt$$

where

$$C_s(x) = \sum_{n>0} \frac{\cos(2\pi nx)}{n^s}$$

is a *Clausen-type* function. For $s = 2k$, even integer, this evaluates to

$$C_{2k}(x) = \frac{(2\pi)^{2k}}{(-1)^{k-1} 2(2k)!} B_{2k}(x)$$

where B_k is a Bernoulli polynomial.

Obviously this also extends to reworkings of (6.6). For example, the $N = 2$ case yields

$$4\beta(s + \frac{1}{2}, \frac{1}{2})\zeta(2s - 1) + \frac{16\pi^{1+s}}{\Gamma(s+1)} \sum_{n=1}^\infty \frac{\sigma_{2s-1}(n)}{n^{s-\frac{3}{2}}} K_{s+\frac{1}{2}}(2n\pi) = 4\zeta(s)\beta(s).$$

This in turn, with $s = 2$, becomes

$$4\pi^3 \sum_{n=1}^\infty \sigma_3(n) e^{-2n\pi} \left(1 + \frac{3}{2} \frac{1}{n\pi} + \frac{3}{4} \frac{1}{n^2\pi^2}\right) \frac{1}{n} = \frac{2}{3} \pi^2 G - \frac{3}{2} \zeta(3),$$

where $G := \sum_{n \geq 0} (-1)^n (2n+1)^{-2}$ is *Catalan's constant*.

There is a puissant formula for θ_2^3 due to Andrews [1] (given with a typographical error in [2] p. 286). It is

$$(6.7) \quad \theta_2^3(q) = 8 \sum_{n=0}^\infty \sum_{j=0}^{2n} \left(\frac{1+q^{4n+2}}{1-q^{4n+2}} \right) q^{(2n+1)^2 - (j+1/2)^2}.$$

Lamentably we have not been able to use it to study R_3 , or even L_3 any further than was achieved in [4].

6.2. Twelve and Twenty-four Squares. Explicit ‘divisor’ formulae for $r_{12}(n)$ and $r_{24}(n)$ are also known (e.g. p. 200 of [15] and §9 of Chapter 9 in [13]): they are

$$r_{12}(n) = 8(-1)^{n-1} \sum_{d|n} (-1)^{d+n/d} d^5 + 16\omega(n)$$

and

$$r_{24}(n) = \frac{16}{691} \sigma_{11}^*(n) + \frac{128}{691} \left((-1)^{n-1} 259\tau(n) - 512\tau\left(\frac{1}{2}n\right) \right)$$

where $\sigma_{11}^*(n) = \sum_{d|n} d^{11}$ if n is odd and $\sigma_{11}^*(n) = \sum_{d|n} (-1)^d d^{11}$ if n is even,

$$q((1-q^2)(1-q^4)(1-q^6)\cdots)^{12} = \sum_{n=1}^{\infty} \omega(n)q^n$$

and

$$q((1-q)(1-q^2)(1-q^3)\cdots)^{24} = \sum_{n=1}^{\infty} \tau(n)q^n.$$

We have recorded these representations because, while $N = 12$ and $N = 24$ (due to Ramanujan, see Chapter IX of [11]) are the next most accessible even cases, neither directly lead to an appropriate closed form for L_N let alone for R_N . This is thanks to the impediment offered by ω and τ respectively: which encode knowledge, via the Jacobi triple-product, of all the representations of n as a sum of 4 or 8 squares. The divisor functions do produce appropriate L-function representations. Thus, using Ramanujan’s ζ -function

$$g_{24}(s) := \sum_{n=1}^{\infty} \frac{\tau(n)}{n^s} = \prod_p (1 - \tau(p)p^{-s} + p^{11-2s})^{-1},$$

which is discussed in detail in Chapter X of [11], it transpires that τ is multiplicative, with the preceding lovely Euler product. Additionally,

$$\begin{aligned} L_{24}(s) = \sum_{n=1}^{\infty} \frac{r_{24}(n)}{n^s} &= \frac{16}{691} (2^{12-2s} - 2^{1-s} + 1) \zeta(s) \zeta(s-11) \\ &+ \frac{128}{691} (259 + 745 \cdot 2^{4-s} + 259 \cdot 2^{12-2s}) g_{24}(s). \end{aligned}$$

Similarly with $g_{12}(s) := \sum_{n=1}^{\infty} \frac{\omega(n)}{n^s}$ one has

$$L_{12}(s) = \sum_{n=1}^{\infty} \frac{r_{12}(n)}{n^s} = 8(1 - 2^{6-2s}) \zeta(s) \zeta(s-5) + 16g_{12}(s).$$

We also note that the analysis in [11] provides an ‘almost closed form’ for

$$\sum_{n=1}^{\infty} \frac{\tau^2(n)}{n^s} = \prod_p \left(1 + \tau^2(p)p^{-s} - p^{22-2s} - \frac{2\tau^2(p)p^{-s}}{1 + p^{11-s}} \right)^{-1}.$$

Finally, we note that a recent paper by Ewell [8] has a new divisor like recursion for τ .

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